NETFLIX ANALYSIS

PROBLEM STATEMENT

With the rapid expansion of digital streaming platforms,Netflix has developed a vast library of movies and Tv shows across different countries,types and durations.However,understanding the patterns within this large dataset is essential to identify trends in content production,audience targeting and platform growth.

This project aims to analayse the Netflix dataset to explore the distribution and characteristics of content available on the platform.The study focuses on identifying patterns in content type(Movies v/s Tv show),release trends over the years,country wise contributions,rating classification and duration.

ABOUT THE DATASET

- The dataset used in this project is the Netflix Dataset. It contains information about movies and TV shows on Netflix.There are about 8,807 records in the dataset.
- It includes details like title, type, director, cast, country, release year, rating, duration, and genre.Some data was missing, so it was cleaned.
- The duration column was also corrected. Movies are measured in minutes, and TV shows are measured in seasons.
- This dataset helps us understand Netflix content, such as popular genres, release trends, and audience types. It is useful for learning data analysis.

AIM OF THE PROJECT

The aim of this dataset is to understand Netflix content. It helps to analyze movies and TV shows, find popular genres, study release year trends, and understand audience ratings. It is mainly used to learn data analysis and visualization.

DAY-1

Importing Libraries

In [1]:
```
import pandas as pd   #data handling
import matplotlib.pyplot as plt   #visualization
import seaborn as sns  #visualization
```

DAY-2

Loading Dataset

In [2]:
```
df=pd.read_csv("netflix.csv")
print(df)
```

```
     show_id     type             movie name         director  \
0         s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1         s2  TV Show          Blood & Water          unknown
2         s3  TV Show              Ganglands  Julien Leclercq
3         s4  TV Show  Jailbirds New Orleans          unknown
4         s5  TV Show           Kota Factory          unknown
...      ...     ...                    ...              ...
8785   s8803    Movie                 Zodiac    David Fincher
8786   s8804  TV Show            Zombie Dumb          unknown
8787   s8805    Movie             Zombieland  Ruben Fleischer
8788   s8806    Movie                   Zoom     Peter Hewitt
8789   s8807    Movie                 Zubaan      Mozez Singh

                                                 cast        country  \
0                                             unknown  United States
1     Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...   South Africa
2     Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...        unknown
3                                             unknown        unknown
4     Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...          India
...                                                 ...            ...
8785  Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...  United States
8786                                          unknown        unknown
8787  Jesse Eisenberg, Woody Harrelson, Emma Stone, ...  United States
8788  Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...  United States
8789  Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...          India

      date_added  release_year rating   duration  \
0     2021-09-25          2020  PG-13     90 min
1     2021-09-24          2021  TV-MA  2 Seasons
2     2021-09-24          2021  TV-MA   1 Season
3     2021-09-24          2021  TV-MA   1 Season
4     2021-09-24          2021  TV-MA  2 Seasons
...          ...           ...    ...        ...
8785  2019-11-20          2007      R    158 min
8786  2019-07-01          2018  TV-Y7  2 Seasons
8787  2019-11-01          2009      R     88 min
8788  2020-01-11          2006     PG     88 min
8789  2019-03-02          2015  TV-14    111 min

                                           listed_in  \
0                                       Documentaries
1       International TV Shows, TV Dramas, TV Mysteries
2     Crime TV Shows, International TV Shows, TV Act...
3                                Docuseries, Reality TV
4     International TV Shows, Romantic TV Shows, TV ...
...                                                 ...
8785                    Cult Movies, Dramas, Thrillers
8786             Kids' TV, Korean TV Shows, TV Comedies
8787                          Comedies, Horror Movies
8788                    Children & Family Movies, Comedies
8789      Dramas, International Movies, Music & Musicals

                                                summary rating_category
0     As her father nears the end of his life, filmm...            Teen
1     After crossing paths at a party, a Cape Town t...           Adult
2     To protect his family from a powerful drug lor...           Adult
3     Feuds, flirtations and toilet talk go down amo...           Adult
4     In a city of coaching centers known to train I...           Adult
...                                                 ...             ...
8785  A political cartoonist, a crime reporter and a...           Adult
8786  While living alone in a spooky town, a young g...            Kids
8787  Looking to survive in a world taken over by zo...           Adult
8788  Dragged from civilian life, a former superhero...          Family
8789  A scrappy but poor boy worms his way into a ty...            Teen

[8790 rows x 13 columns]
```

Inspect Data

In [3]:
```
df.head(10)
```

Out[3]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	Teen
1	s2	TV Show	Blood & Water	unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	Adult
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	Adult
3	s4	TV Show	Jailbirds New Orleans	unknown	unknown	unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	Adult
4	s5	TV Show	Kota Factory	unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train i...	Adult
5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	unknown	2021-09-24	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries	The arrival of a charismatic young priest brin...	Adult
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	unknown	2021-09-24	2021	PG	91 min	Children & Family Movies	Equestria's divided. But a bright-eyed hero be...	Family
7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies	On a photo shoot in Ghana, an American model s...	Adult
8	s9	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-24	2021	TV-14	9 Seasons	British TV Shows, Reality TV	A talented batch of amateur bakers face off in...	Teen
9	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	2021-09-24	2021	PG-13	104 min	Comedies, Dramas	A woman adjusting to life after a loss contend...	Teen

In [4]:
```
df.tail(10)
```

Out[4]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
8780	s8798	TV Show	Zak Storm	unknown	Michael Johnston, Jessica Gee-George, Christin...	United States, France, South Korea, Indonesia	2018-09-13	2016	TV-Y7	3 Seasons	Kids' TV	Teen surfer Zak Storm is mysteriously transpor...	Kids
8781	s8799	Movie	Zed Plus	Chandra Prakash Dwivedi	Adil Hussain, Mona Singh, K.K. Raina, Sanjay M...	India	2019-12-31	2014	TV-MA	131 min	Comedies, Dramas, International Movies	A philandering small-town mechanic's political...	Adult
8782	s8800	Movie	Zenda	Avadhoot Gupte	Santosh Juvekar, Siddharth Chandekar, Sachit P...	India	2018-02-15	2009	TV-14	120 min	Dramas, International Movies	A change in the leadership of a political part...	Teen
8783	s8801	TV Show	Zindagi Gulzar Hai	unknown	Sanam Saeed, Fawad Khan, Ayesha Omer, Mehreen ...	Pakistan	2016-12-15	2012	TV-PG	1 Season	International TV Shows, Romantic TV Shows, TV ...	Strong-willed, middle-class Kashaf and carefre...	Family
8784	s8802	Movie	Zinzana	Majid Al Ansari	Ali Suliman, Saleh Bakri, Yasa, Ali Al-Jabri, ...	United Arab Emirates, Jordan	2016-03-09	2015	TV-MA	96 min	Dramas, International Movies, Thrillers	Recovering alcoholic Talal wakes up inside a s...	Adult
8785	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	Adult
8786	s8804	TV Show	Zombie Dumb	unknown	unknown	unknown	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	Kids
8787	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	Adult
8788	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	Family
8789	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	Teen

In [5]:
```
df.sample(20)
```

Out[5]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
4313	s4314	Movie	Out of Many, One	John Hoffman, Nanfu Wang	unknown	United States	2018-12-12	2018	TV-PG	35 min	Documentaries	In this documentary short, a diverse group of ...	Family
4663	s4664	TV Show	Quantico	unknown	Priyanka Chopra, Josh Hopkins, Jake McLaughlin...	United States	2018-09-02	2018	TV-14	3 Seasons	Crime TV Shows, TV Dramas, TV Mysteries	When evidence in a deadly terrorist attack imp...	Teen
6799	s6808	Movie	From Japan to Egypt	Mahmoud Karim	Ahmed Eid, Saki Tsukamoto, Nada Moussa, Mohamm...	Egypt	2019-06-06	2017	TV-14	93 min	Children & Family Movies, Comedies, Dramas	After his wife relocates to her home country o...	Teen
7932	s7949	TV Show	Save Our Shelter	unknown	Rocky Kanaka, Rob North	United States	2018-08-18	2015	TV-G	1 Season	Reality TV	Pet rescue expert Rocky Kanaka and renovation ...	Family
4213	s4214	Movie	El Potro: Unstoppable	Lorena Munoz	Jimena Barón, Rodrigo Romero, Diego Cremonesi,...	unknown	2019-01-04	2018	TV-MA	122 min	Dramas, International Movies, Music & Musicals	Argentine cuarteto singer Rodrigo "El Potro" B...	Adult
6923	s6934	Movie	Harry and Snowman	Ron Davis	Harry DeLeyer	United States	2017-01-15	2015	TV-PG	83 min	Documentaries, Sports Movies	With only a strong will and a cheap horse, a f...	Family
4445	s4446	Movie	Kai Po Che!	Abhishek Kapoor	Rajkummar Rao, Amit Sadh, Sushant Singh Rajput...	India	2018-11-01	2013	TV-MA	120 min	Dramas, International Movies, Sports Movies	This Bollywood drama tells the story of three ...	Adult
5053	s5054	Movie	Newness	Drake Doremus	Nicholas Hoult, Laia Costa, Danny Huston, Cour...	United States	2018-02-01	2017	TV-MA	117 min	Dramas, Independent Movies, Romantic Movies	Martin and Gabi try to form a relationship aft...	Adult
8216	s8234	Movie	The Bygone	Graham Phillips, Parker Phillips	Graham Phillips, Shawn Hatosy, Jamie McShane, ...	United States	2020-03-23	2019	TV-MA	106 min	Independent Movies, Thrillers	His way of life under threat due to the oil fr...	Adult
2277	s2278	Movie	Madras Café	Shoojit Sircar	John Abraham, Nargis Fakhri, Rashi Khanna, Avi...	India	2020-07-05	2013	TV-14	128 min	Dramas, International Movies, Thrillers	An Indian intelligence agent journeys to a war...	Teen
4551	s4552	TV Show	Monty Python's Flying Circus	unknown	John Cleese, Graham Chapman, Terry Jones, Eric...	United Kingdom	2018-10-02	1974	TV-14	4 Seasons	British TV Shows, Classic & Cult TV, Internati...	The Monty Python players make their mark with ...	Teen
6188	s6195	TV Show	Atomic Puppet	unknown	Eric Bauza, Lisa Norton, Carlos Díaz, Peter Ol...	Canada, France	2017-12-01	2016	TV-Y7	1 Season	Crime TV Shows, Kids' TV, TV Comedies	Captain Atomic – once a superhero, now a sock ...	Kids
1789	s1790	Movie	Secrets of the Saqqara Tomb	James Tovell	unknown	United States	2020-10-28	2020	TV-PG	114 min	Documentaries	After unearthing a tomb that had been untouche...	Family
7637	s7653	Movie	On Yoga The Architecture of Peace	Heitor Dhalia	Michael O'Neill	Brazil, India, China, United States	2018-01-10	2017	TV-14	92 min	Documentaries, Faith & Spirituality	Based on photographer Michael O'Neill's book o...	Teen
7553	s7569	TV Show	Nature: Animals With Cameras	unknown	Gordon Buchanan	United Kingdom, United States	2018-10-01	2018	TV-PG	1 Season	Docuseries, Science & Nature TV	With custom-built cameras attached to their bo...	Family
79	s80	Movie	Tughlaq Durbar (Telugu)	Delhiprasad Deenadayalan	Vijay Sethupathi, Parthiban, Raashi Khanna	unknown	2021-09-11	2021	TV-14	145 min	Comedies, Dramas, International Movies	A budding politician has devious plans to rise...	Teen
6174	s6181	Movie	Apollo 18	Gonzalo López-Gallego	Warren Christie, Lloyd Owen, Ryan Robbins, Mic...	United States, Canada	2019-03-09	2011	PG-13	87 min	Horror Movies, Sci-Fi & Fantasy, Thrillers	This sci-fi thriller suggests that Apollo 17, ...	Teen
1982	s1983	Movie	Anaamika	Sekhar Kammula	Nayantara, Vaibhav Reddy, Pasupathy, Harshvard...	India	2020-09-17	2014	TV-14	130 min	International Movies, Thrillers	As a woman scours Hyderabad for her missing hu...	Teen
1107	s1108	Movie	God Calling	BB Sasore	Zainab Balogun, Karibi Fubara, Diana Egwuatu, ...	Nigeria	2021-04-02	2018	TV-MA	120 min	Dramas, Faith & Spirituality, International Mo...	The loss of a child plunges Sade into a suicid...	Adult
5747	s5749	Movie	Bleach The Movie: Fade to Black	Noriyuki Abe	Masakazu Morita, Fumiko Orikasa, Aya Hirano, H...	Japan	2016-10-15	2008	TV-PG	94 min	Action & Adventure, Anime Features, Sci-Fi & F...	When his mind is fogged by a scimitar-weilding...	Family

---

Check Column Data Type

In [6]:
```
df.dtypes
```

Out[6]:
```
show_id            object
type               object
movie name         object
director           object
cast               object
country            object
date_added         object
release_year        int64
rating             object
duration           object
listed_in          object
summary            object
rating_category    object
dtype: object
```

Display Column Names

In [7]:
```
df.columns
```

Out[7]:
```
Index(['show_id', 'type', 'movie name', 'director', 'cast', 'country',
       'date_added', 'release_year', 'rating', 'duration', 'listed_in',
       'summary', 'rating_category'],
      dtype='object')
```

Display Number Of Rows And Columns

In [8]:
```
df.shape
```

Out[8]:
```
(8790, 13)
```

In [9]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8790 entries, 0 to 8789
Data columns (total 13 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   show_id          8790 non-null   object
 1   type             8790 non-null   object
 2   movie name       8790 non-null   object
 3   director         8790 non-null   object
 4   cast             8790 non-null   object
 5   country          8790 non-null   object
 6   date_added       8790 non-null   object
 7   release_year     8790 non-null   int64
 8   rating           8790 non-null   object
 9   duration         8790 non-null   object
 10  listed_in        8790 non-null   object
 11  summary          8790 non-null   object
 12  rating_category  8790 non-null   object
dtypes: int64(1), object(12)
memory usage: 892.9+ KB
```

- total 12 columns
- memory usage: 825.8+ KB
- missing values(director)=2634
- missing values(cast)=825
- missing values(country):831
- missing values(data_added):10
- missing values(ratings):4
- missing values(duration):3
- data type mistakes(duration):duration is not object,it is int64

---

selecting columns

In [10]:
```
df["type"]
```

Out[10]:
```
0         Movie
1       TV Show
2       TV Show
3       TV Show
4       TV Show
          ...
8785      Movie
8786    TV Show
8787      Movie
8788      Movie
8789      Movie
Name: type, Length: 8790, dtype: object
```

In [11]:
```
df["duration"]
```

Out[11]:
```
0          90 min
1       2 Seasons
2        1 Season
3        1 Season
4       2 Seasons
          ...
8785     158 min
8786   2 Seasons
8787      88 min
8788      88 min
8789     111 min
Name: duration, Length: 8790, dtype: object
```

In [12]:
```
df["rating"]
```

Out[12]:
```
0       PG-13
1       TV-MA
2       TV-MA
3       TV-MA
4       TV-MA
        ...
8785        R
8786    TV-Y7
8787        R
8788       PG
8789    TV-14
Name: rating, Length: 8790, dtype: object
```

In [13]:
```
df["country"]
```

Out[13]:
```
0       United States
1        South Africa
2             unknown
3             unknown
4               India
            ...
8785    United States
8786          unknown
8787    United States
8788    United States
8789            India
Name: country, Length: 8790, dtype: object
```

Selecting Multiple Columns

In [14]:
```
df[["type","country"]]
```

Out[14]:

	type	country
0	Movie	United States
1	TV Show	South Africa
2	TV Show	unknown
3	TV Show	unknown
4	TV Show	India
...	...	...
8785	Movie	United States
8786	TV Show	unknown
8787	Movie	United States
8788	Movie	United States
8789	Movie	India

8790 rows × 2 columns

In [15]:
```
new=df[["country","cast"]]
```

In [16]:
```
new
```

Out[16]:

	country	cast
0	United States	unknown
1	South Africa	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...
2	unknown	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...
3	unknown	unknown
4	India	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...
...	...	...
8785	United States	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...
8786	unknown	unknown
8787	United States	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...
8788	United States	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...
8789	India	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...

8790 rows × 2 columns

Filter The Data

In [17]:
```
df[df["release_year"]<1950]
```

Out[17]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
1331	s1332	TV Show	Five Came Back: The Reference Films	unknown	unknown	United States	2021-02-09	1945	TV-MA	1 Season	TV Shows	This collection includes 12 World War II-era p...	Adult
4250	s4251	TV Show	Pioneers: First Women Filmmakers*	unknown	unknown	unknown	2018-12-30	1925	TV-14	1 Season	TV Shows	This collection restores films from women who ...	Teen
7208	s7220	Movie	Know Your Enemy - Japan	Frank Capra, Joris Ivens	Walter Huston, Dana Andrews	United States	2017-03-31	1945	TV-14	63 min	Classic Movies, Documentaries	Though culturally insensitive by modern standa...	Teen
7282	s7295	Movie	Let There Be Light	John Huston	Walter Huston	United States	2017-03-31	1946	TV-PG	58 min	Classic Movies, Documentaries	Some returning combat veterans suffer scars th...	Family
7560	s7576	Movie	Nazi Concentration Camps	George Stevens	unknown	United States	2017-03-31	1945	TV-MA	59 min	Classic Movies, Documentaries	Shocking footage shows Nazi concentration camp...	Adult
7728	s7744	TV Show	Pioneers of African-American Cinema	Oscar Micheaux, Spencer Williams, Richard E. N...	unknown	United States	2017-02-01	1946	TV-14	1 Season	TV Shows	This newly preserved collection features more ...	Teen
7775	s7791	Movie	Prelude to War	Frank Capra	unknown	United States	2017-03-31	1942	TV-14	52 min	Classic Movies, Documentaries	Frank Capra's documentary chronicles the rise ...	Teen
7914	s7931	Movie	San Pietro	John Huston	unknown	United States	2017-03-31	1945	TV-14	32 min	Classic Movies, Documentaries	After the Allies invade Italy, the Liri Valley...	Teen
8188	s8206	Movie	The Battle of Midway	John Ford	Henry Fonda, Jane Darwell	United States	2017-03-31	1942	TV-14	18 min	Classic Movies, Documentaries	Director John Ford captures combat footage of ...	Teen
8402	s8420	Movie	The Memphis Belle: A Story of a\nFlying Fortress	William Wyler	unknown	United States	2017-03-31	1944	TV-PG	40 min	Classic Movies, Documentaries	This documentary centers on the crew of the B-...	Family
8419	s8437	Movie	The Negro Soldier	Stuart Heisler	unknown	United States	2017-03-31	1944	TV-14	40 min	Classic Movies, Documentaries	This documentary urged African Americans to en...	Teen
8570	s8588	Movie	Thunderbolt	William Wyler, John Sturges	James Stewart, Lloyd Bridges	United States	2017-03-31	1947	TV-PG	42 min	Classic Movies, Documentaries	A P-47 Thunderbolt squadron is shown in prepar...	Family
8623	s8641	Movie	Tunisian Victory	Frank Capra, John Huston, Hugh Stewart, Roy Bo...	Burgess Meredith	United States, United Kingdom	2017-03-31	1944	TV-14	76 min	Classic Movies, Documentaries	British and American troops join forces to lib...	Teen
8643	s8661	Movie	Undercover: How to Operate Behind Enemy Lines	John Ford	unknown	United States	2017-03-31	1943	TV-PG	61 min	Classic Movies, Documentaries	This World War II-era training film dramatizes...	Family
8722	s8740	Movie	Why We Fight: The Battle of Russia	Frank Capra, Anatole Litvak	unknown	United States	2017-03-31	1943	TV-PG	82 min	Documentaries	This installment of Frank Capra's acclaimed do...	Family
8746	s8764	Movie	WWII: Report from the Aleutians	John Huston	unknown	United States	2017-03-31	1943	TV-PG	45 min	Documentaries	Filmmaker John Huston narrates this Oscar-nomi...	Family

In [18]:
```
df[df["type"]=="Movie"]
```

Out[18]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	Teen
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	unknown	2021-09-24	2021	PG	91 min	Children & Family Movies	Equestria's divided. But a bright-eyed hero be...	Family
7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies	On a photo shoot in Ghana, an American model s...	Adult
9	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	2021-09-24	2021	PG-13	104 min	Comedies, Dramas	A woman adjusting to life after a loss contend...	Teen
12	s13	Movie	Je Suis Karl	Christian Schwochow	Luna Wedler, Jannis Niewöhner, Milan Peschel, ...	Germany, Czech Republic	2021-09-23	2021	TV-MA	127 min	Dramas, International Movies	After most of her family is murdered in a terr...	Adult
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8784	s8802	Movie	Zinzana	Majid Al Ansari	Ali Suliman, Saleh Bakri, Yasa, Ali Al-Jabri, ...	United Arab Emirates, Jordan	2016-03-09	2015	TV-MA	96 min	Dramas, International Movies, Thrillers	Recovering alcoholic Talal wakes up inside a s...	Adult
8785	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	Adult
8787	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	Adult
8788	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	Family
8789	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	Teen

6126 rows × 13 columns

In [19]:
```
df[df["country"]=="United States"]
```

Out[19]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	Teen
9	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	2021-09-24	2021	PG-13	104 min	Comedies, Dramas	A woman adjusting to life after a loss contend...	Teen
15	s16	TV Show	Dear White People	unknown	Logan Browning, Brandon P. Bell, DeRon Horton,...	United States	2021-09-22	2021	TV-MA	4 Seasons	TV Comedies, TV Dramas	Students of color navigate the daily slights a...	Adult
27	s28	Movie	Grown Ups	Dennis Dugan	Adam Sandler, Kevin James, Chris Rock, David S...	United States	2021-09-20	2010	PG-13	103 min	Comedies	Mourning the loss of their beloved junior high...	Teen
28	s29	Movie	Dark Skies	Scott Stewart	Keri Russell, Josh Hamilton, J.K. Simmons, Dak...	United States	2021-09-19	2013	PG-13	97 min	Horror Movies, Sci-Fi & Fantasy	A family's idyllic suburban life shatters when...	Teen
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8774	s8792	Movie	Young Adult	Jason Reitman	Charlize Theron, Patton Oswalt, Patrick Wilson...	United States	2019-11-20	2011	R	94 min	Comedies, Dramas, Independent Movies	When a divorced writer gets a letter from an o...	Adult
8776	s8794	Movie	Yours, Mine and Ours	Raja Gosnell	Dennis Quaid, Rene Russo, Sean Faris, Katija P...	United States	2019-11-20	2005	PG	88 min	Children & Family Movies, Comedies	When a father of eight and a mother of 10 prep...	Family
8785	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	Adult
8787	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	Adult
8788	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	Family

2809 rows × 13 columns

In [20]:
```
df[df["director"]=="Peter Hewitt"]
```

Out[20]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
8788	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	Family

In [21]:
```
df[df["release_year"]>2000]
```

Out[21]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	Teen
1	s2	TV Show	Blood & Water	unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	Adult
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	Adult
3	s4	TV Show	Jailbirds New Orleans	unknown	unknown	unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	Adult
4	s5	TV Show	Kota Factory	unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	Adult
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8785	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	Adult
8786	s8804	TV Show	Zombie Dumb	unknown	unknown	unknown	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	Kids
8787	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	Adult
8788	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	Family
8789	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	Teen

8228 rows × 13 columns

In [22]:
```
df[df["listed_in"]=="Dramas"]
```

Out[22]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
112	s113	Movie	Worth	Sara Colangelo	Michael Keaton, Stanley Tucci, Amy Ryan, Shuno...	unknown	2021-09-03	2021	PG-13	119 min	Dramas	In the wake of the Sept. 11 attacks, a lawyer ...	Teen
129	s130	Movie	An Unfinished Life	Lasse Hallström	Robert Redford, Jennifer Lopez, Morgan Freeman...	Germany, United States	2021-09-01	2005	PG-13	108 min	Dramas	A grieving widow and her daughter move in with...	Teen
142	s143	Movie	Freedom Writers	Richard LaGravenese	Hilary Swank, Patrick Dempsey, Scott Glenn, Im...	Germany, United States	2021-09-01	2007	PG-13	124 min	Dramas	While her at-risk students are reading classic...	Teen
162	s163	Movie	Marshall	Reginald Hudlin	Chadwick Boseman, Josh Gad, Kate Hudson, Sterl...	United States, China, Hong Kong	2021-09-01	2017	PG-13	118 min	Dramas	This biopic of Thurgood Marshall, the first Bl...	Teen
329	s330	Movie	Catch Me If You Can	Steven Spielberg	Leonardo DiCaprio, Tom Hanks, Christopher Walk...	United States, Canada	2021-08-01	2002	PG-13	142 min	Dramas	An FBI agent makes it his mission to put cunni...	Teen
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8657	s8675	Movie	Viceroy's House	Gurinder Chadha	Hugh Bonneville, Gillian Anderson, Manish Daya...	United Kingdom, India, Sweden	2017-12-12	2017	NR	106 min	Dramas	As viceroy Lord Mountbatten arrives in Delhi t...	Unrated
8658	s8676	Movie	Victor	Brandon Dickerson	Patrick Davis, Lisa Vidal, Josh Pence, José Zú...	United States	2017-07-24	2015	PG-13	110 min	Dramas	In 1962 Brooklyn, a Puerto Rican teen who join...	Teen
8668	s8686	Movie	VS.	Ed Lilly	Connor Swindells, Fola Evans-Akingbola, Nichol...	United Kingdom	2019-06-19	2018	TV-MA	99 min	Dramas	A young man in foster care finds his voice in ...	Adult
8674	s8692	Movie	Wakefield	Robin Swicord	Bryan Cranston, Jennifer Garner, Jason O'Mara,...	United States	2019-03-02	2016	R	109 min	Dramas	An unhappy father and lawyer quits his suburba...	Adult
8679	s8697	Movie	War Horse	Steven Spielberg	Emily Watson, David Thewlis, Peter Mullan, Nie...	United States, India	2019-05-06	2011	PG-13	147 min	Dramas	During World War I, the bond between a young E...	Teen

137 rows × 13 columns

Filter with multiple condition

In [23]:
```
df[(df["country"]=="India")&(df["release_year"]<2000)]
```

Out[23]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
24	s25	Movie	Jeans	S. Shankar	Prashanth, Aishwarya Rai Bachchan, Sri Lakshmi...	India	2021-09-21	1998	TV-14	166 min	Comedies, International Movies, Romantic Movies	When the father of the man she loves insists t...	Teen
114	s115	Movie	Anjaam	Rahul Rawail	Madhuri Dixit, Shah Rukh Khan, Tinnu Anand, Jo...	India	2021-09-02	1994	TV-14	143 min	Dramas, International Movies, Thrillers	A wealthy industrialist's dangerous obsession ...	Teen
253	s254	Movie	Bombay	Mani Ratnam	Arvind Swamy, Manisha Koirala, Nasser, Raja Kr...	India	2021-08-18	1995	TV-14	135 min	Dramas, International Movies, Romantic Movies	Years after they elope against their family's ...	Teen
542	s543	Movie	Ujala	Naresh Saigal	Mala Sinha, Shammi Kapoor, Raaj Kumar, Leela C...	India	2021-07-06	1959	TV-14	143 min	Dramas, International Movies	An honest man dreams of a better life for his ...	Teen
1441	s1442	Movie	Hello Brother	Sohail Khan	Salman Khan, Arbaaz Khan, Rani Mukerji, Johny ...	India	2021-01-04	1999	TV-14	126 min	Action & Adventure, Comedies, International Mo...	After receiving a heart transplant from a murd...	Teen
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8120	s8137	Movie	Swami	Basu Chatterjee	Shabana Azmi, Girish Karnad, Vikram, Dheeraj K...	India	2019-12-31	1977	TV-PG	123 min	Dramas, Independent Movies, International Movies	After she enters into an arranged marriage, a ...	Family
8128	s8145	Movie	Taal	Subhash Ghai	Anil Kapoor, Aishwarya Rai Bachchan, Akshaye K...	India	2018-03-01	1999	TV-PG	177 min	Dramas, International Movies, Music & Musicals	Businessman Manav falls for aspiring singer Ma...	Family
8315	s8333	Movie	The Great Gambler	Shakti Samanta	Amitabh Bachchan, Zeenat Aman, Neetu Singh, Pr...	India	2019-12-31	1979	TV-14	161 min	Action & Adventure, Classic Movies, Cult Movies	After unwittingly using his skills to obtain m...	Teen
8611	s8629	Movie	Trikal (Past, Present, Future)	Shyam Benegal	Leela Naidu, Neena Gupta, Anita Kanwar, Soni R...	India	2019-12-31	1985	TV-14	134 min	Comedies, Dramas, Independent Movies	A Goan native returns to his now-derelict, pal...	Teen
8612	s8630	Movie	Trimurti	Mukul Anand	Anil Kapoor, Shah Rukh Khan, Jackie Shroff, Sa...	India	2018-03-01	1995	TV-PG	173 min	Action & Adventure, Dramas, International Movies	After being framed, a policewoman languishing ...	Family

81 rows × 13 columns

Selecting Rows(iloc method)

In [ ]:

In [24]:
```
df.iloc[150]
```

Out[24]:
```
show_id                                                 s151
type                                                    Movie
movie name                                        In Too Deep
director                                       Michael Rymer
cast            Omar Epps, LL Cool J, Nia Long, Stanley Tucci,...
country                                        United States
date_added                                        2021-09-01
release_year                                            1999
rating                                                     R
duration                                             97 min
listed_in                                          Thrillers
summary         Rookie cop Jeffrey Cole poses as a drug dealer...
rating_category                                        Adult
Name: 150, dtype: object
```

In [25]:
```
df.iloc[100:200]
```

Out[25]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
100	s101	TV Show	Tobot Galaxy Detectives	unknown	Austin Abell, Travis Turner, Cole Howard, Anna...	unknown	2021-09-07	2019	TV-Y7	2 Seasons	Kids' TV	An intergalactic device transforms toy cars in...	Kids
101	s102	Movie	Untold: Breaking Point	Chapman Way, Maclain Way	unknown	United States	2021-09-07	2021	TV-MA	80 min	Documentaries, Sports Movies	Under pressure to continue a winning tradition...	Adult
102	s103	TV Show	Countdown: Inspiration4 Mission to Space	Jason Hehir	unknown	unknown	2021-09-06	2021	TV-14	1 Season	Docuseries, Science & Nature TV	From training to launch to landing, this all-a...	Teen
103	s104	Movie	Shadow Parties	Yemi Amodu	Jide Kosoko, Omotola Jalade-Ekeinde, Yemi Blaq...	unknown	2021-09-06	2020	TV-MA	117 min	Dramas, International Movies, Thrillers	A family faces destruction in a long-running c...	Adult
104	s105	TV Show	Tayo the Little Bus	unknown	Robyn Slade, Kami Desilets	South Korea	2021-09-06	2016	TV-Y	2 Seasons	Kids' TV, Korean TV Shows	As they learn their routes around the busy cit...	Kids
...	...	...	...	...	...	...	...	...	...	...	...	...	...
195	s196	Movie	EMI: Liya Hai To Chukana Padega	Saurabh Kabra	Sanjay Dutt, Arjun Rampal, Malaika Arora, Aash...	India	2021-08-27	2008	TV-14	128 min	Comedies, Dramas, International Movies	A bank hires an enigmatic and unorthodox debt ...	Teen
196	s197	Movie	He's All That	Mark Waters	Addison Rae, Tanner Buchanan, Rachael Leigh Co...	unknown	2021-08-27	2021	TV-14	92 min	Comedies, Romantic Movies	An influencer specializing in makeovers bets s...	Teen
197	s198	TV Show	I Heart Arlo	unknown	Michael J. Woodard, Mary Lambert, Jonathan Van...	unknown	2021-08-27	2021	TV-Y7	1 Season	Kids' TV, TV Comedies	It's a whole new world for Arlo and his one-of...	Kids
198	s199	TV Show	King of Boys: The Return of the King	Kemi Adetiba	Sola Sobowale, Toni Tones, Richard Mofe-Damijo...	Nigeria	2021-08-27	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Dramas	Alhaja Eniola Salami starts anew and sets her ...	Adult
199	s200	Movie	Koi Aap Sa	Partho Mitra	Aftab Shivdasani, Natassha, Dipannita Sharma, ...	India	2021-08-27	2006	TV-14	135 min	Comedies, Dramas, International Movies	Star athlete Rohan has his eye on a beautiful ...	Teen

100 rows × 13 columns

Selecting Rows(loc method)

In [26]:
```
df.loc[500,"type"]
```

Out[26]:
```
'Movie'
```

Rename Column

In [27]:
```
df.rename(columns={"title":"movie name","description":"summary"},inplace="true")
```

In [28]:
```
df.iloc[100:105]
```

Out[28]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
100	s101	TV Show	Tobot Galaxy Detectives	unknown	Austin Abell, Travis Turner, Cole Howard, Anna...	unknown	2021-09-07	2019	TV-Y7	2 Seasons	Kids' TV	An intergalactic device transforms toy cars in...	Kids
101	s102	Movie	Untold: Breaking Point	Chapman Way, Maclain Way	unknown	United States	2021-09-07	2021	TV-MA	80 min	Documentaries, Sports Movies	Under pressure to continue a winning tradition...	Adult
102	s103	TV Show	Countdown: Inspiration4 Mission to Space	Jason Hehir	unknown	unknown	2021-09-06	2021	TV-14	1 Season	Docuseries, Science & Nature TV	From training to launch to landing, this all-a...	Teen
103	s104	Movie	Shadow Parties	Yemi Amodu	Jide Kosoko, Omotola Jalade-Ekeinde, Yemi Blaq...	unknown	2021-09-06	2020	TV-MA	117 min	Dramas, International Movies, Thrillers	A family faces destruction in a long-running c...	Adult
104	s105	TV Show	Tayo the Little Bus	unknown	Robyn Slade, Kami Desilets	South Korea	2021-09-06	2016	TV-Y	2 Seasons	Kids' TV, Korean TV Shows	As they learn their routes around the busy cit...	Kids

Value Count

In [29]:
```
df["type"].value_counts()
```

Out[29]:
```
type
Movie      6126
TV Show    2664
Name: count, dtype: int64
```

In [30]:
```
df["country"].value_counts()
```

Out[30]:
```
country
United States                              2809
India                                       972
unknown                                     829
United Kingdom                              418
Japan                                       243
                                           ...
Romania, Bulgaria, Hungary                    1
Uruguay, Guatemala                            1
France, Senegal, Belgium                      1
Mexico, United States, Spain, Colombia        1
United Arab Emirates, Jordan                  1
Name: count, Length: 749, dtype: int64
```

DAY-3

Data Cleaning

Clean Numerical Columns With Characters

In [31]:
```
# df["engine"]=pd.to_numeric(
#    .astype(str)
#    .str.replace(r'[^0]','',regax=True)
#    .replace('',None),
#    error='coerce'
#)
```

Parse Date_added As Datetime

In [32]:
```
df["date_added"]=pd.to_datetime(df["date_added"],
                                errors="coerce")
```

In [33]:
```
df.dtypes
```

Out[33]:
```
show_id                    object
type                       object
movie name                 object
director                   object
cast                       object
country                    object
date_added         datetime64[ns]
release_year                int64
rating                     object
duration                   object
listed_in                  object
summary                    object
rating_category            object
dtype: object
```

In [34]:
```
df['rating'].value_counts()
```

Out[34]:
```
rating
TV-MA       3205
TV-14       2157
TV-PG        861
R            799
PG-13        490
TV-Y7        333
TV-Y         306
PG           287
TV-G         220
NR            79
G             41
TV-Y7-FV       6
NC-17          3
UR             3
Name: count, dtype: int64
```

In [35]:
```
import pandas as pd

def categorize_rating(rating):
    if pd.isna(rating):
        return "Unknown"

    rating = str(rating).strip()

    # Handle wrong entries like durations
    if "min" in rating:
        return "Unknown"

    # Kids content
    if rating in ["TV-Y", "TV-Y7", "TV-Y7-FV", "G"]:
        return "Kids"

    # Family / General
    elif rating in ["TV-G", "PG", "TV-PG"]:
        return "Family"

    # Teens
    elif rating in ["PG-13", "TV-14"]:
        return "Teen"

    # Adults
    elif rating in ["R", "TV-MA", "NC-17"]:
        return "Adult"

    # Not Rated / Others
```

```
    elif rating in ["NR", "UR"]:
        return "Unrated"

    else:
        return "Unrated"
```

In [36]:
```
df["rating_category"] = df["rating"].apply(categorize_rating)
```

In [37]:
```
print(df["rating_category"].value_counts())
```

```
rating_category
Adult      4007
Teen       2647
Family     1368
Kids        686
Unrated      82
Name: count, dtype: int64
```

In [38]:
```
df
```

Out[38]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	Teen
1	s2	TV Show	Blood & Water	unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	Adult
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	Adult
3	s4	TV Show	Jailbirds New Orleans	unknown	unknown	unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	Adult
4	s5	TV Show	Kota Factory	unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	Adult
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8785	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	Adult
8786	s8804	TV Show	Zombie Dumb	unknown	unknown	unknown	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	Kids
8787	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	Adult
8788	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	Family
8789	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	Teen

8790 rows × 13 columns

In [39]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8790 entries, 0 to 8789
Data columns (total 13 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   show_id          8790 non-null   object
 1   type             8790 non-null   object
 2   movie name       8790 non-null   object
 3   director         8790 non-null   object
 4   cast             8790 non-null   object
 5   country          8790 non-null   object
 6   date_added       8790 non-null   datetime64[ns]
 7   release_year     8790 non-null   int64
 8   rating           8790 non-null   object
 9   duration         8790 non-null   object
 10  listed_in        8790 non-null   object
 11  summary          8790 non-null   object
 12  rating_category  8790 non-null   object
dtypes: datetime64[ns](1), int64(1), object(11)
memory usage: 892.9+ KB
```

Handling Missing Values

In [40]:
```
df.isnull()
```

Out[40]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
0	False	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8785	False	False	False	False	False	False	False	False	False	False	False	False	False
8786	False	False	False	False	False	False	False	False	False	False	False	False	False
8787	False	False	False	False	False	False	False	False	False	False	False	False	False
8788	False	False	False	False	False	False	False	False	False	False	False	False	False
8789	False	False	False	False	False	False	False	False	False	False	False	False	False

8790 rows × 13 columns

In [41]:
```
df.isnull().sum()
```

Out[41]:
```
show_id            0
type               0
movie name         0
director           0
cast               0
country            0
date_added         0
release_year       0
rating             0
duration           0
listed_in          0
summary            0
rating_category    0
dtype: int64
```

How To Fill Missing Values

In [42]:
```
df['director'].fillna("unknown",inplace=True)
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_13860\2513338713.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df['director'].fillna("unknown",inplace=True)
```

In [43]:
```
df['country'].fillna("unknown",inplace=True)
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_13860\367023652.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df['country'].fillna("unknown",inplace=True)
```

In [44]:
```
df['cast'].fillna("unknown",inplace=True)
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_13860\2462058862.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df['cast'].fillna("unknown",inplace=True)
```

In [45]:
```
df.dropna(subset=["date_added","rating","duration"],inplace=True)
```

In [46]:
```
df.isnull().sum()
```

Out[46]:
```
show_id            0
type               0
movie name         0
director           0
cast               0
country            0
date_added         0
release_year       0
rating             0
duration           0
listed_in          0
summary            0
rating_category    0
dtype: int64
```

In [47]:
```
df.shape
```

Out[47]:
```
(8790, 13)
```

Handling Duplicates

In [48]:
```
df.duplicated().sum()
```

Out[48]:
```
np.int64(0)
```

In [49]:
```
# print("before:",df.shape)
# df=df.drop_duplicates()
# print("after:",df.shape)
```

DAY-4

Ensuring Data Cleaning

In [50]:
```
df.head(10)
```

Out[50]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	Teen
1	s2	TV Show	Blood & Water	unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	Adult
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	Adult
3	s4	TV Show	Jailbirds New Orleans	unknown	unknown	unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	Adult
4	s5	TV Show	Kota Factory	unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	Adult
5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	unknown	2021-09-24	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries	The arrival of a charismatic young priest brin...	Adult
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	unknown	2021-09-24	2021	PG	91 min	Children & Family Movies	Equestria's divided. But a bright-eyed hero be...	Family
7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies	On a photo shoot in Ghana, an American model s...	Adult
8	s9	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-24	2021	TV-14	9 Seasons	British TV Shows, Reality TV	A talented batch of amateur bakers face off in...	Teen
9	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	2021-09-24	2021	PG-13	104 min	Comedies, Dramas	A woman adjusting to life after a loss contend...	Teen

In [51]:
```
df.isnull().sum()
```

Out[51]:
```
show_id            0
type               0
movie name         0
director           0
cast               0
country            0
date_added         0
release_year       0
rating             0
duration           0
listed_in          0
summary            0
rating_category    0
dtype: int64
```

In [52]:
```
df.columns
```

Out[52]:
```
Index(['show_id', 'type', 'movie name', 'director', 'cast', 'country',
       'date_added', 'release_year', 'rating', 'duration', 'listed_in',
       'summary', 'rating_category'],
      dtype='object')
```

In [53]:
```
df.dtypes
```

Out[53]:
```
show_id                    object
type                       object
movie name                 object
director                   object
cast                       object
country                    object
date_added         datetime64[ns]
release_year               int64
rating                     object
duration                   object
listed_in                  object
summary                    object
rating_category            object
dtype: object
```

In [54]:
```
df.sample(10)
```

Out[54]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
5193	s5194	TV Show	The Little Nyonya	unknown	Jeanette Aw, Qi Yu Wu, Pierre Png, Dai Xiangyu...	unknown	2017-11-01	2009	TV-14	1 Season	International TV Shows, TV Dramas	An orphaned Chinese woman in Singapore refuses...	Teen
7649	s7665	Movie	One More Shot	Noah Moskin	unknown	United States	2018-01-15	2017	TV-MA	89 min	Documentaries	Infertility affects millions. One couple copes...	Adult
2221	s2222	Movie	Jack Whitehall: I'm Only Joking	Dave Skinner, Freddie Waters	Jack Whitehall	United Kingdom	2020-07-21	2020	TV-MA	58 min	Stand-Up Comedy	Jack Whitehall hits the stage with hilarious t...	Adult
5303	s5304	Movie	Manoranjan	Shammi Kapoor	Sanjeev Kumar, Zeenat Aman, Shammi Kapoor, Dev...	India	2017-09-01	1974	TV-14	162 min	Comedies, International Movies, Music & Musicals	After getting fired, a police officer connects...	Teen
6409	s6416	Movie	Candyman	Bernard Rose	Virginia Madsen, Tony Todd, Xander Berkeley, K...	United States, United Kingdom	2019-10-01	1992	R	99 min	Cult Movies, Horror Movies	Grad student Helen Lyle unintentionally summon...	Adult
7574	s7590	TV Show	Ninja Turtles: The Next Mutation	unknown	Jarred Blancard, Mitchell A. Lee Yuen, Dean Ch...	Canada, United States	2016-01-01	1997	TV-G	1 Season	Kids' TV, TV Comedies	Everyone's favorite teenage mutants are back i...	Family
1931	s1932	Movie	The Boys in the Band	Joe Mantello	Jim Parsons, Zachary Quinto, Matt Bomer, Andre...	United States	2020-09-30	2020	R	122 min	Dramas, LGBTQ Movies	At a birthday party in 1968 New York, a surpri...	Adult
6607	s6614	TV Show	Dino Hunt	unknown	unknown	Canada	2017-08-04	2016	TV-G	1 Season	Docuseries, Science & Nature TV	Top scientists and paleontologists examine din...	Family
2348	s2349	Movie	The Royal Hibiscus Hotel	Ishaya Bako	Zainab Balogun, Kenneth Okolie, Deyemi Okanlaw...	Nigeria	2020-06-22	2017	TV-PG	90 min	Comedies, International Movies, Romantic Movies	A disenchanted chef tries to help her parents ...	Family
2905	s2906	Movie	MR. RIGHT	Paco Cabezas	Sam Rockwell, Anna Kendrick, Tim Roth, James R...	United States	2020-02-16	2015	R	96 min	Action & Adventure, Comedies, Romantic Movies	Fresh off a bad relationship, thrill-seeking M...	Adult

In [55]:
```
df.shape
```

Out[55]:
```
(8790, 13)
```

In [56]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8790 entries, 0 to 8789
Data columns (total 13 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   show_id          8790 non-null   object
 1   type             8790 non-null   object
 2   movie name       8790 non-null   object
 3   director         8790 non-null   object
 4   cast             8790 non-null   object
 5   country          8790 non-null   object
 6   date_added       8790 non-null   datetime64[ns]
 7   release_year     8790 non-null   int64
 8   rating           8790 non-null   object
 9   duration         8790 non-null   object
 10  listed_in        8790 non-null   object
 11  summary          8790 non-null   object
 12  rating_category  8790 non-null   object
dtypes: datetime64[ns](1), int64(1), object(11)
memory usage: 892.9+ KB
```

Filtering The Data -movie and tv show

In [57]:
```
movies=df[df['type']=='Movie']
```

In [58]:
```
movies
```

Out[58]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	Teen
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	unknown	2021-09-24	2021	PG	91 min	Children & Family Movies	Equestria's divided. But a bright-eyed hero be...	Family
7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies	On a photo shoot in Ghana, an American model s...	Adult
9	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	2021-09-24	2021	PG-13	104 min	Comedies, Dramas	A woman adjusting to life after a loss contend...	Teen
12	s13	Movie	Je Suis Karl	Christian Schwochow	Luna Wedler, Jannis Niewöhner, Milan Peschel, ...	Germany, Czech Republic	2021-09-23	2021	TV-MA	127 min	Dramas, International Movies	After most of her family is murdered in a terr...	Adult
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8784	s8802	Movie	Zinzana	Majid Al Ansari	Ali Suliman, Saleh Bakri, Yasa, Ali Al-Jabri, ...	United Arab Emirates, Jordan	2016-03-09	2015	TV-MA	96 min	Dramas, International Movies, Thrillers	Recovering alcoholic Talal wakes up inside a s...	Adult
8785	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	Adult
8787	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	Adult
8788	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	Family
8789	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	Teen

6126 rows × 13 columns

Parase Movies As Numerical

In [59]:
```
movies["duration"] = movies["duration"].str.replace("min"," ",regex=False)
movies["duration"] = pd.to_numeric(movies["duration"], errors="coerce")
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_13860\1304505552.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  movies["duration"] = movies["duration"].str.replace("min"," ",regex=False)
C:\Users\HP\AppData\Local\Temp\ipykernel_13860\1304505552.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  movies["duration"] = pd.to_numeric(movies["duration"], errors="coerce")
```

In [60]:
```
movies.dtypes
```

Out[60]:
```
show_id                    object
type                       object
movie name                 object
director                   object
cast                       object
country                    object
date_added        datetime64[ns]
release_year               int64
rating                     object
duration                   int64
listed_in                  object
summary                    object
rating_category            object
dtype: object
```

In [61]:
```
tv_shows=df[df['type']=='TV Show']
```

In [62]:
```
tv_shows
```

Out[62]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
1	s2	TV Show	Blood & Water	unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	Adult
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	Adult
3	s4	TV Show	Jailbirds New Orleans	unknown	unknown	unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	Adult
4	s5	TV Show	Kota Factory	unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	Adult
5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	unknown	2021-09-24	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries	The arrival of a charismatic young priest brin...	Adult
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8778	s8796	TV Show	Yu-Gi-Oh! Arc-V	unknown	Mike Liscio, Emily Bauer, Billy Bob Thompson, ...	Japan, Canada	2018-05-01	2015	TV-Y7	2 Seasons	Anime Series, Kids' TV	Now that he's discovered the Pendulum Summonin...	Kids
8779	s8797	TV Show	Yunus Emre	unknown	Gökhan Atalay, Payidar Tüfekçioglu, Baran Akbu...	Turkey	2017-01-17	2016	TV-PG	2 Seasons	International TV Shows, TV Dramas	During the Mongol invasions, Yunus Emre leaves...	Family
8780	s8798	TV Show	Zak Storm	unknown	Michael Johnston, Jessica Gee-George, Christin...	United States, France, South Korea, Indonesia	2018-09-13	2016	TV-Y7	3 Seasons	Kids' TV	Teen surfer Zak Storm is mysteriously transpor...	Kids
8783	s8801	TV Show	Zindagi Gulzar Hai	unknown	Sanam Saeed, Fawad Khan, Ayesha Omer, Mehreen ...	Pakistan	2016-12-15	2012	TV-PG	1 Season	International TV Shows, Romantic TV Shows, TV ...	Strong-willed, middle-class Kashaf and carefre...	Family
8786	s8804	TV Show	Zombie Dumb	unknown	unknown	unknown	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	Kids

2664 rows × 13 columns

Parase TV_show To Numeric

In [63]:
```
tv_shows["duration"] = tv_shows["duration"].astype(str)
tv_shows["duration"] = tv_shows["duration"].str.split().str[0].astype(int)
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_13860\4025971363.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  tv_shows["duration"] = tv_shows["duration"].astype(str)
C:\Users\HP\AppData\Local\Temp\ipykernel_13860\4025971363.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  tv_shows["duration"] = tv_shows["duration"].str.split().str[0].astype(int)
```

In [64]:
```
tv_shows
```

Out[64]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
1	s2	TV Show	Blood & Water	unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	Adult
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	unknown	2021-09-24	2021	TV-MA	1	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	Adult
3	s4	TV Show	Jailbirds New Orleans	unknown	unknown	unknown	2021-09-24	2021	TV-MA	1	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	Adult
4	s5	TV Show	Kota Factory	unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	Adult
5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	unknown	2021-09-24	2021	TV-MA	1	TV Dramas, TV Horror, TV Mysteries	The arrival of a charismatic young priest brin...	Adult
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8778	s8796	TV Show	Yu-Gi-Oh! Arc-V	unknown	Mike Liscio, Emily Bauer, Billy Bob Thompson, ...	Japan, Canada	2018-05-01	2015	TV-Y7	2	Anime Series, Kids' TV	Now that he's discovered the Pendulum Summonin...	Kids
8779	s8797	TV Show	Yunus Emre	unknown	Gökhan Atalay, Payidar Tüfekçioglu, Baran Akbu...	Turkey	2017-01-17	2016	TV-PG	2	International TV Shows, TV Dramas	During the Mongol invasions, Yunus Emre leaves...	Family
8780	s8798	TV Show	Zak Storm	unknown	Michael Johnston, Jessica Gee-George, Christin...	United States, France, South Korea, Indonesia	2018-09-13	2016	TV-Y7	3	Kids' TV	Teen surfer Zak Storm is mysteriously transpor...	Kids
8783	s8801	TV Show	Zindagi Gulzar Hai	unknown	Sanam Saeed, Fawad Khan, Ayesha Omer, Mehreen ...	Pakistan	2016-12-15	2012	TV-PG	1	International TV Shows, Romantic TV Shows, TV ...	Strong-willed, middle-class Kashaf and carefre...	Family
8786	s8804	TV Show	Zombie Dumb	unknown	unknown	unknown	2019-07-01	2018	TV-Y7	2	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	Kids

2664 rows × 13 columns

Save To File

In [65]:
```
df.to_csv("netflix_cleaned.csv",index=False)
```

In [66]:
```
movies.to_csv("movies.csv",index=False)
```

In [67]:
```
tv_shows.to_csv("tv_shows.csv",index=False)
```

Load New Data

In [68]:
```
df_new=pd.read_csv("netflix_cleaned.csv")
```

In [69]:
```
df_new
```

Out[69]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	Teen
1	s2	TV Show	Blood & Water	unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	Adult
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	Adult
3	s4	TV Show	Jailbirds New Orleans	unknown	unknown	unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	Adult
4	s5	TV Show	Kota Factory	unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	Adult
...	...	...	...	...	...	...	...	...	...	...	...	...	...
8785	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	Adult
8786	s8804	TV Show	Zombie Dumb	unknown	unknown	unknown	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	Kids
8787	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	Adult
8788	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	Family
8789	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	Teen

8790 rows × 13 columns

In [70]:
```
df_new1=pd.read_csv("movies.csv")
```

In [71]:
```
df_new1
```

Out[71]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	unknown	United States	2021-09-25	2020	PG-13	90	Documentaries	As her father nears the end of his life, filmm...	Teen
1	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	unknown	2021-09-24	2021	PG	91	Children & Family Movies	Equestria's divided. But a bright-eyed hero be...	Family
2	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125	Dramas, Independent Movies, International Movies	On a photo shoot in Ghana, an American model s...	Adult
3	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	2021-09-24	2021	PG-13	104	Comedies, Dramas	A woman adjusting to life after a loss contend...	Teen
4	s13	Movie	Je Suis Karl	Christian Schwochow	Luna Wedler, Jannis Niewöhner, Milan Peschel, ...	Germany, Czech Republic	2021-09-23	2021	TV-MA	127	Dramas, International Movies	After most of her family is murdered in a terr...	Adult
...	...	...	...	...	...	...	...	...	...	...	...	...	...
6121	s8802	Movie	Zinzana	Majid Al Ansari	Ali Suliman, Saleh Bakri, Yasa, Ali Al-Jabri, ...	United Arab Emirates, Jordan	2016-03-09	2015	TV-MA	96	Dramas, International Movies, Thrillers	Recovering alcoholic Talal wakes up inside a s...	Adult
6122	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...	Adult
6123	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88	Comedies, Horror Movies	Looking to survive in a world taken over by zo...	Adult
6124	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...	Family
6125	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...	Teen

6126 rows × 13 columns

In [72]:
```
df_new2=pd.read_csv("tv_shows.csv")
```

In [73]:
```
df_new2
```

Out[73]:

	show_id	type	movie name	director	cast	country	date_added	release_year	rating	duration	listed_in	summary	rating_category
0	s2	TV Show	Blood & Water	unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	Adult
1	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	unknown	2021-09-24	2021	TV-MA	1	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	Adult
2	s4	TV Show	Jailbirds New Orleans	unknown	unknown	unknown	2021-09-24	2021	TV-MA	1	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	Adult
3	s5	TV Show	Kota Factory	unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...	Adult
4	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	unknown	2021-09-24	2021	TV-MA	1	TV Dramas, TV Horror, TV Mysteries	The arrival of a charismatic young priest brin...	Adult
...	...	...	...	...	...	...	...	...	...	...	...	...	...
2659	s8796	TV Show	Yu-Gi-Oh! Arc-V	unknown	Mike Liscio, Emily Bauer, Billy Bob Thompson, ...	Japan, Canada	2018-05-01	2015	TV-Y7	2	Anime Series, Kids' TV	Now that he's discovered the Pendulum Summonin...	Kids
2660	s8797	TV Show	Yunus Emre	unknown	Gökhan Atalay, Payidar Tüfekçioglu, Baran Akbu...	Turkey	2017-01-17	2016	TV-PG	2	International TV Shows, TV Dramas	During the Mongol invasions, Yunus Emre leaves...	Family
2661	s8798	TV Show	Zak Storm	unknown	Michael Johnston, Jessica Gee-George, Christin...	United States, France, South Korea, Indonesia	2018-09-13	2016	TV-Y7	3	Kids' TV	Teen surfer Zak Storm is mysteriously transpor...	Kids
2662	s8801	TV Show	Zindagi Gulzar Hai	unknown	Sanam Saeed, Fawad Khan, Ayesha Omer, Mehreen ...	Pakistan	2016-12-15	2012	TV-PG	1	International TV Shows, Romantic TV Shows, TV ...	Strong-willed, middle-class Kashaf and carefre...	Family
2663	s8804	TV Show	Zombie Dumb	unknown	unknown	unknown	2019-07-01	2018	TV-Y7	2	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...	Kids

2664 rows × 13 columns

Summary Statistics

Average(mean)

In [74]:
```
print("release_year  -",df_new["release_year"].mean())
```

```
release_year  - 2014.183162684869
```

Median

In [75]:
```
print("release_year  -",df_new["release_year"].median())
```

```
release_year  - 2017.0
```

Mode

In [76]:
```
df_new["release_year"].mode()
```

Out[76]:
```
0    2018
Name: release_year, dtype: int64
```

In [77]:
```
print("release_year  -",df_new["release_year"].mode())
```

```
release_year  - 0    2018
Name: release_year, dtype: int64
```

Standard Deviation

In [78]:
```
print("release_year  -",df_new["release_year"].std())
```

```
release_year  - 8.82546602834746
```

In [79]:
```
df_new.select_dtypes(include='number').mean()
```

Out[79]:
```
release_year    2014.183163
dtype: float64
```

In [80]:
```
df_new.select_dtypes(include='number').median()
```

Out[80]:
```
release_year    2017.0
dtype: float64
```

In [81]:
```
df_new.select_dtypes(include='object').mode()
```

Out[81]:

	show_id	type	movie name	director	cast	country	date_added	rating	duration	listed_in	summary	rating_category
0	s1	Movie	#Alive	unknown	unknown	United States	2020-01-01	TV-MA	1 Season	Dramas, International Movies	Paranormal activity at a lush, abandoned prope...	Adult
1	s10	NaN	#AnneFrank - Parallel Stories	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	s100	NaN	#FriendButMarried	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	s1000	NaN	#FriendButMarried 2	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	s1001	NaN	#Roxy	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...
8785	s995	NaN	忍者ハットリくん	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8786	s996	NaN	海的儿子	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8787	s997	NaN	마녀사냥	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8788	s998	NaN	반드시 잡는다	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8789	s999	NaN	최강전사 미니특공대 : 영웅의 탄생	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

8790 rows × 12 columns

In [82]:
```
df_new.select_dtypes(include='number').std()
```

Out[82]:
```
release_year    8.825466
dtype: float64
```

In [83]:
```
df_new.describe()
```

Out[83]:

	release_year
count	8790.000000
mean	2014.183163
std	8.825466
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

- the data set contains 8790 entries
- the average release year is 2014
- the standard deviation is 8.825
- the oldest content dates back to 1925
- 25% of the content was released before 2013
- The median release year is 2017
- 75% of the content was released before 2019
- The latest content is from 2021

In [84]:
```
df_new.describe(include='object')
```

Out[84]:

	show_id	type	movie name	director	cast	country	date_added	rating	duration	listed_in	summary	rating_category
count	8790	8790	8790	8790	8790	8790	8790	8790	8790	8790	8790	8790
unique	8790	2	8790	4527	7679	749	1713	14	220	513	8758	5
top	s1	Movie	Dick Johnson Is Dead	unknown	unknown	United States	2020-01-01	TV-MA	1 Season	Dramas, International Movies	Paranormal activity at a lush, abandoned prope...	Adult
freq	1	6126	1	2621	825	2809	110	3205	1791	362	4	4007

- total 8790 entries,means no missing values
- 2 unique values are movie and tv shows,dataset is heavily movie-dominate
- every titles are unique
- 4527 directors are unique ,the most of them are unknown(2621)
- 7679 cast are unique,most of them are unknown(825)
- most content dominated by the united states
- 110 shows were released on 01/01/2020
- majority content TV-MA
- drama & international movies are the majority of the listed_in
- most target on adult audience

In [85]:
```
df_new1.describe(include='object')
```

Out[85]:

	show_id	type	movie name	director	cast	country	date_added	rating	listed_in	summary	rating_category
count	6126	6126	6126	6126	6126	6126	6126	6126	6126	6126	6126
unique	6126	1	6126	4353	5444	652	1531	14	278	6100	5
top	s1	Movie	Dick Johnson Is Dead	unknown	unknown	United States	2020-01-01	TV-MA	Dramas, International Movies	Paranormal activity at a lush, abandoned prope...	Adult
freq	1	6126	1	187	475	2055	97	2062	362	4	2862

- Total 6126 entries,means no missing
- only movies are in the dstaset
- 4353 directors are unique,187 directors are unknown
- 475 cast are unknown
- most of the movies dominated by united status
- 97 movies are released on 01/01/2020
- majority content is TV-MA
- dramas & international Movies are the majority of the listed_in
- 205 movies are unique duration,152 movies are 90 min duration

In [86]:
```
df_new2.describe(include='object')
```

Out[86]:

	show_id	type	movie name	director	cast	country	date_added	rating	listed_in	summary	rating_category
count	2664	2664	2664	2664	2664	2664	2664	2664	2664	2664	2664
unique	2664	1	2664	223	2272	197	1012	9	235	2660	5
top	s2	TV Show	Blood & Water	unknown	unknown	United States	2021-07-06	TV-MA	Kids' TV	As a psychology professor faces Alzheimer's, h...	Adult
freq	1	2664	1	2434	350	754	31	1143	219	2	1145

- total 2664 entries,means no missing
- only tv show are in the data set
- 223 directors are unique,2434 directors are unknown
- 350 cast are unknown
- most of the tv shows dominated by united states
- 31 movies are released on 06/07/2021
- majority content is TV-MA
- 15 tv shows are unique,1791 tv shows are 1 season
- kid's tv are the majority of the listed_in

DAY-5

Data Visualization

Univariate Analysis

- histogram,kde,box plot - numerical
- count, pie - categorical

Bivariate Analysis

- bar plot - category vs numerical
- line plot(time series) and scatter - numerical vs numerical

Multivariate Analysis

- heat map - correlation
- pair plot - numerical

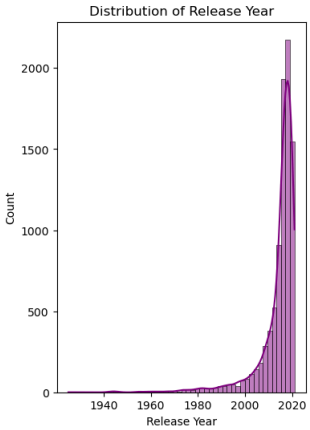
Univariate Analysis

Histogram

- Distribution of continous numerical data

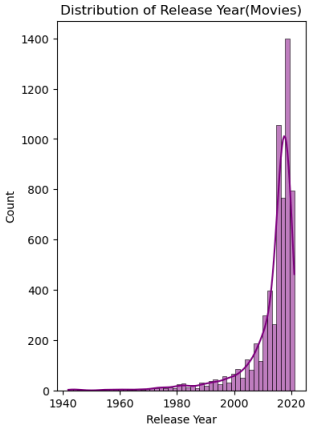
In [87]:
```
import seaborn as sns
import matplotlib.pyplot as plt
```

In [88]:
```
plt.figure(figsize=(4,6))
sns.histplot(data=df_new,x="release_year",bins=50,color="purple",kde=True)
plt.title("Distribution of Release Year")
plt.xlabel("Release Year")
plt.show()
```



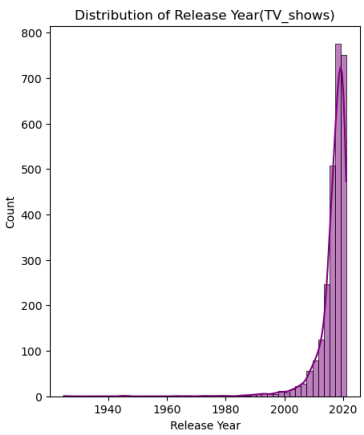
- The histogram shows the distribution of release year
- A huge majority of the releases happened in the last 10-15 years.
- There are very few releases from before the year 2000.
- The no.of releases started growing very fast after 2010.
- 2018 and 2019 were the biggest year of releases.
- The numbers drop suddenly after 2019
- The distribution is strongly left-skewed
- Most of the data is crowded on the right side (recent years)

In [89]:
```
plt.figure(figsize=(4,6))
sns.histplot(data=df_new1,x="release_year",bins=50,color="purple",kde=True)
plt.title("Distribution of Release Year(Movies)")
plt.xlabel("Release Year")
plt.show()
```



- Most content is released after 2000
- Majority of release happened in 10-15 years
- Releases from 1940 to 1980 are very rare
- There is a massive spike right around 2018
- After 2018 peak,there is a quick decline in the no.of releases in leading into 2020

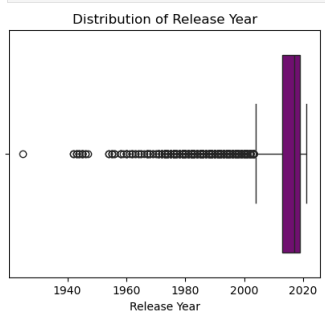
In [90]:
```
plt.figure(figsize=(5,6))
sns.histplot(data=df_new2,x="release_year",bins=50,color="purple",kde=True)
plt.title("Distribution of Release Year(TV_shows)")
plt.xlabel("Release Year")
plt.show()
```



- Almost all tv shows were released after 2010
- TV shows from 1940 to 2000 are extremely rare
- 2019 has the most releases followed by a small drop in 2020
- Majority of release happened in last 5-10 years

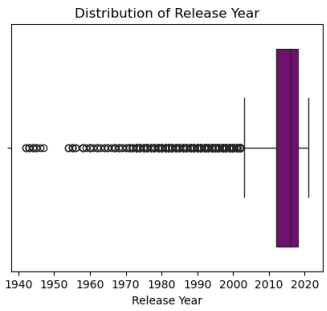
Box Plot

In [91]:
```
plt.figure(figsize=(5,4))
sns.boxplot(data=df_new,x="release_year",color="purple")
plt.title("Distribution of Release Year")
plt.xlabel("Release Year")
plt.show()
```



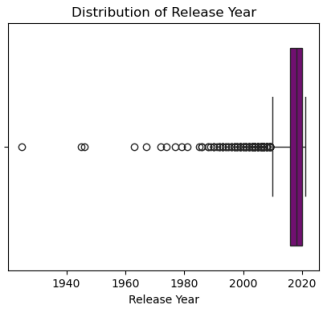
- 50% of all releases happened between approximately 2013 and 2019
- The vertical line in the box shows the middle value is around 2017
- The long trail of circles to the left shows that releases before 2005 are rare and considered outliers

In [92]:
```
plt.figure(figsize=(5,4))
sns.boxplot(data=df_new1,x="release_year",color="purple")
plt.title("Distribution of Release Year")
plt.xlabel("Release Year")
plt.show()
```



- Most data is concentrated in 2010–2020
- The distribution is skewed toward recent years
- Many older years appear as outliers
- The median is in recent years
- Very few entries before 2000

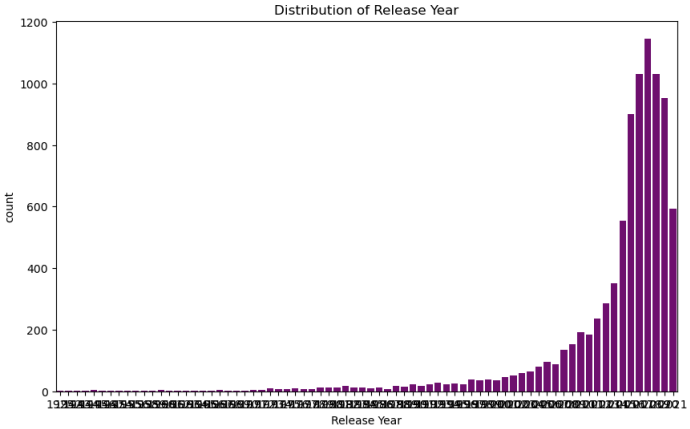
In [93]:
```
plt.figure(figsize=(5,4))
sns.boxplot(data=df_new2,x="release_year",color="purple")
plt.title("Distribution of Release Year")
plt.xlabel("Release Year")
plt.show()
```



- Most titles are recent: the catalog is heavily weighted toward releases after 2010.
- Movies dominate the dataset over TV shows.
- Adult-rated content is the largest rating group.
- Netflix content peaks around 2018–2019 and drops after 2019.
- TV shows are usually short-run (1–3 seasons) in this dataset.
- Older releases (pre-2000) are rare and appear as outliers.

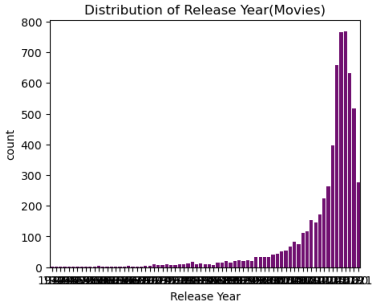
Count Plot

In [94]:
```
plt.figure(figsize=(10,6))
sns.countplot(data=df_new,x="release_year",color="purple")
plt.title("Distribution of Release Year")
plt.xlabel("Release Year")
plt.show()
```



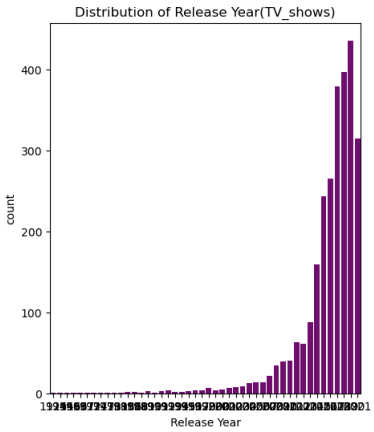
- Most releaes are after 2010
- there is only very few releases before 2000
- Release increases quickly after 2010
- The highest releases count are in 2018–2019
- After 2019, counts drop a bit
- The data is right-heavy, meaning recent years dominate

In [95]:
```
plt.figure(figsize=(5,4))
sns.countplot(data=df_new1,x="release_year",color="purple")
plt.title("Distribution of Release Year(Movies)")
plt.xlabel("Release Year")
plt.show()
```



- Most titles are from 2010 onward.
- Movies far outnumber TV shows.
- Adult-rated content is the largest group.
- Release volume peaks around 2018–2019.

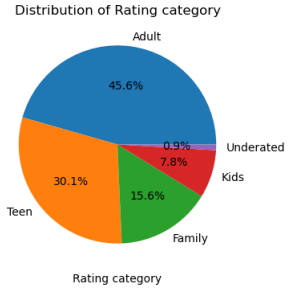
In [96]:
```
plt.figure(figsize=(5,6))
sns.countplot(data=df_new2,x="release_year",color="purple")
plt.title("Distribution of Release Year(TV_shows)")
plt.xlabel("Release Year")
plt.show()
```



- Most catalog entries are recent (after 2010).
- Movies dominate over TV shows.
- Adult-rated content is the largest group.

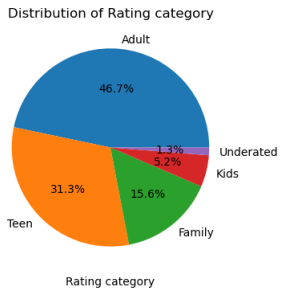
Pie Chart

In [97]:
```
x=df_new['rating_category'].value_counts()
labels=['Adult','Teen','Family','Kids','Underated']
plt.figure(figsize=(5,4))
plt.pie(x,labels=labels,autopct='%1.1f%%')
plt.title("Distribution of Rating category")
plt.xlabel("Rating category")
plt.show()
```

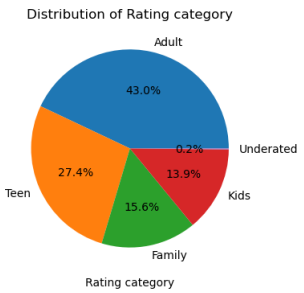


- This pie chart shows how many titles are in each rating group.
- **Adult** is the biggest group, so most titles are for mature viewers.
- **Teen** is the next biggest group.
- **Family** and **Kids** are smaller, so child-friendly content is less common.
- **Unrated** titles are very few.

In [98]:
```
x=df_new1['rating_category'].value_counts()
labels=['Adult','Teen','Family','Kids','Underated']
plt.figure(figsize=(5,4))
plt.pie(x,labels=labels,autopct='%1.1f%%')
plt.title("Distribution of Rating category")
plt.xlabel("Rating category")
plt.show()
```



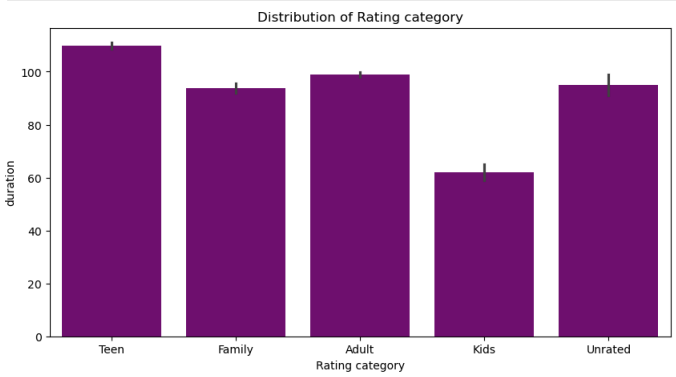
In [99]:
```
x=df_new2['rating_category'].value_counts()
labels=['Adult','Teen','Family','Kids','Underated']
plt.figure(figsize=(5,4))
plt.pie(x,labels=labels,autopct='%1.1f%%')
plt.title("Distribution of Rating category")
plt.xlabel("Rating category")
plt.show()
```



Bivariate Analysis

Bar Plot

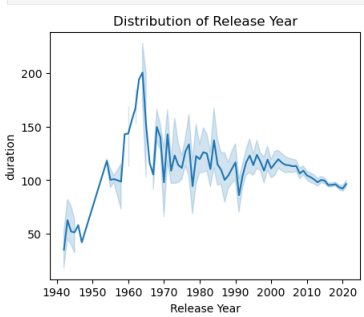
In [100…
```
plt.figure(figsize=(10,5))
sns.barplot(data=df_new1,x="rating_category",y="duration",color="purple")
plt.title("Distribution of Rating category")
plt.xlabel("Rating category")
plt.show()
```



- The bar plot shows average length for each rating group.
- **Teen** and **Adult** categories have the longest durations.
- **Kids** content is much shorter than other groups.

Line Plot

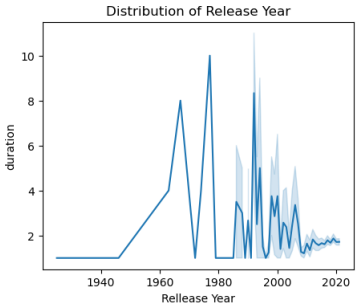
In [101…
```
plt.figure(figsize=(5,4))
sns.lineplot(data=df_new1,x="release_year",y=("duration"))
plt.title("Distribution of Release Year")
plt.xlabel("Release Year")
plt.show()
```



- Movie lengths increased sharply from the 1940s to the mid-1960s
- Peak average duration occurred around the 1960s
- High variability in movie lengths during the 1960s–1980s
- Gradual decline in duration after the 1970s
- Modern films (2000s onward) are shorter and more consistent
- Overall trend: rise → peak → steady decline → stabilization

In [102…

```
plt.figure(figsize=(5,4))
sns.lineplot(data=df_new2,x="release_year",y=("duration"))
plt.title("Distribution of Release Year")
plt.xlabel("Rellease Year")
plt.show()
```

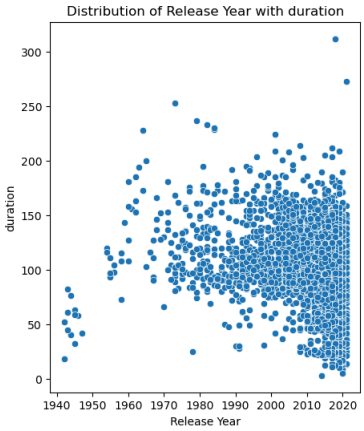


- Values stay very low and flat until around the 1950s
- Sharp spikes appear between the 1960s–1980s (high instability)
- Peak occurs around the late 1970s–early 1980s
- After 1990, values drop and fluctuate frequently
- From 2000 onward, values stabilize at low levels
- Overall trend: flat → volatile spikes → decline → low, stable pattern

Scatter Plot

In [103…

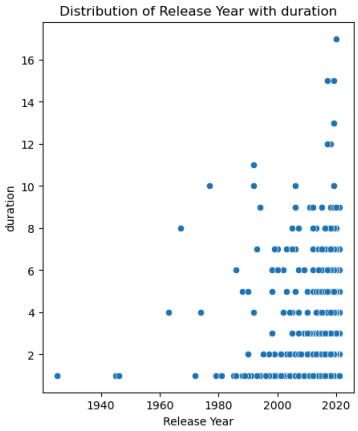
```
plt.figure(figsize=(5,6))
sns.scatterplot(data=df_new1,x="release_year",y=("duration"))
plt.title("Distribution of Release Year with duration")
plt.xlabel("Release Year")
plt.show()
```



- Most movies cluster between ~80–150 minutes
- After 2000, many more movies are released (denser points)
- Wide spread in durations, but majority stay in mid-range
- Some outliers: very short (<50 min) and very long (>200 min) films
- Older films (pre-1970) are fewer but often longer
- Overall: high variation, but typical movie length stays around ~100–140 minutes

In [104…

```
plt.figure(figsize=(5,6))
sns.scatterplot(data=df_new2,x="release_year",y=("duration"))
plt.title("Distribution of Release Year with duration")
plt.xlabel("Release Year")
plt.show()
```



- Most TV shows are of smaller duration(1-3 seasons)
- The number of releases increases very much after 2010.
- Very long-running shows (duration above 10) are rare and appear as outliers.
- The plot suggests Netflix content is skewed toward newer, shorter-run TV = - shows rather than long multi-season series.

DAY-6

Multivariate Analysis

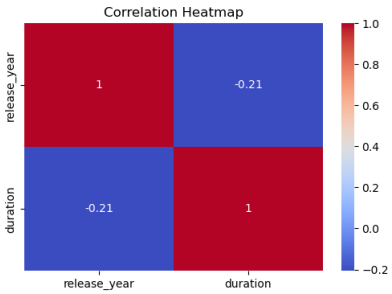
HeatMap

In [105…
```
df_new1_num=df_new1.select_dtypes(include='number')
df_new1_num.corr()
```

Out[105…

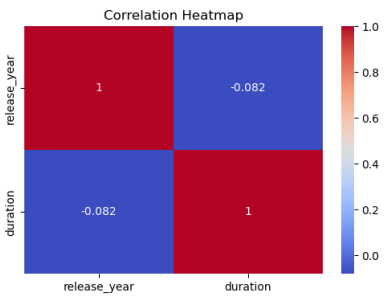
	release_year	duration
release_year	1.000000	-0.206248
duration	-0.206248	1.000000

In [106…
```
plt.figure(figsize=(6,4))
sns.heatmap(df_new1.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



- The correlation between release_year and duration is -0.21.
- This is a weak negative relationship → as release year increases, duration tends to slightly decrease.
- The value is close to 0, so the relationship is not strong.
- This means newer content is only slightly shorter on average, but not by much.
- There is no strong pattern between release year and duration.

In [107…
```
plt.figure(figsize=(6,4))
sns.heatmap(df_new2.corr(numeric_only=True),annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

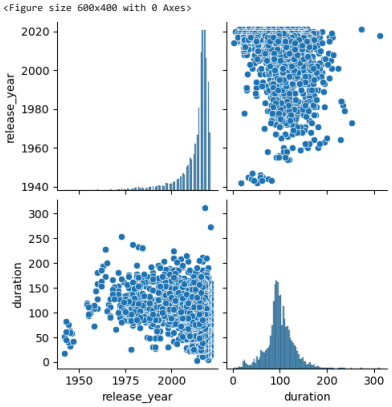


Pair Plot

In [108…
```
plt.figure(figsize=(6,4))
sns.pairplot(df_new1.select_dtypes(include='number'), palette='coolwarm')
```

```
c:\Users\HP\anaconda3\New folder\Lib\site-packages\seaborn\axisgrid.py:1513: UserWarning: Ignoring `palette` because no `hue` variable has been assigned.
  func(x=vector, **plot_kwargs)
c:\Users\HP\anaconda3\New folder\Lib\site-packages\seaborn\axisgrid.py:1513: UserWarning: Ignoring `palette` because no `hue` variable has been assigned.
  func(x=vector, **plot_kwargs)
c:\Users\HP\anaconda3\New folder\Lib\site-packages\seaborn\axisgrid.py:1615: UserWarning: Ignoring `palette` because no `hue` variable has been assigned.
  func(x=x, y=y, **kwargs)
c:\Users\HP\anaconda3\New folder\Lib\site-packages\seaborn\axisgrid.py:1615: UserWarning: Ignoring `palette` because no `hue` variable has been assigned.
  func(x=x, y=y, **kwargs)
```

Out[108…
```
<seaborn.axisgrid.PairGrid at 0x2341832acf0>
<Figure size 600x400 with 0 Axes>
```



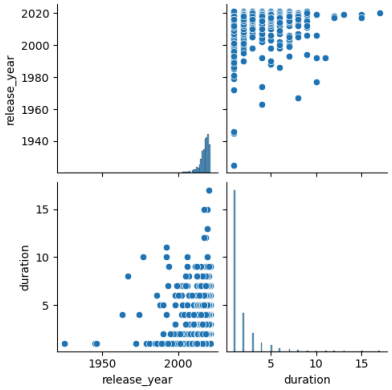
- The pairplot shows no strong linear relationship between release_year and duration.
- Points are scattered widely, so newer movies are not consistently longer or shorter.
- Most movies cluster around typical durations, regardless of release year.
- The trend is almost flat, confirming the weak negative correlation.

In [109…
```
plt.figure(figsize=(6,4))
sns.pairplot(df_new2.select_dtypes(include='number'), palette='coolwarm')
```

```
c:\Users\HP\anaconda3\New folder\Lib\site-packages\seaborn\axisgrid.py:1513: UserWarning: Ignoring `palette` because no `hue` variable has been assigned.
  func(x=vector, **plot_kwargs)
c:\Users\HP\anaconda3\New folder\Lib\site-packages\seaborn\axisgrid.py:1513: UserWarning: Ignoring `palette` because no `hue` variable has been assigned.
  func(x=vector, **plot_kwargs)
c:\Users\HP\anaconda3\New folder\Lib\site-packages\seaborn\axisgrid.py:1615: UserWarning: Ignoring `palette` because no `hue` variable has been assigned.
  func(x=x, y=y, **kwargs)
c:\Users\HP\anaconda3\New folder\Lib\site-packages\seaborn\axisgrid.py:1615: UserWarning: Ignoring `palette` because no `hue` variable has been assigned.
  func(x=x, y=y, **kwargs)
```

Out[109…

```
<seaborn.axisgrid.PairGrid at 0x2341ae1fed0>
<Figure size 600x400 with 0 Axes>
```



- most releases are from 2010 onward.
- peak release activity is around 2018–2019.
- Movies dominate the dataset.TV shows are fewer and mostly short-run.
- Ratings are skewed toward adult content, with teen content second-largest.
- TV show durations are mostly 1–3 seasons; long series are rare.
- Release year vs duration shows only a weak negative correlation.

DAY-7

CONCLUSION

In this project, the Netflix Dataset was successfully cleaned and analyzed. Several changes were made to improve the quality of the data. Missing values in columns like director, cast, and country were handled by filling or removing them.The movies and tv shows are seperated to a file. The duration column was cleaned and converted into a proper numeric format, where movies are shown in minutes and TV shows in seasons.

After cleaning, the dataset became more accurate and easy to use for analysis. The analysis showed that Netflix has more movies than TV shows, most content is released after 2010, and popular genres include drama and comedy. It also showed that content is mainly made for young and adult audiences.

Overall, this project helped in understanding how to clean data, fix errors, and perform data analysis. It shows the importance of data preprocessing before doing any analysis.

KEY INSIGHTS

- Movies dominated in netflix than tv shows.
- In the dataset 6126 Moviesand 2664 Tv shows are there.
- Most content dominated by United States.
- Most of the movies and tv shows target on adult audience.
- After cleaning the dataset ,have 8790 entries.
- The average release year is 2014.
- The median release year is 2017.
- The standard deviation of release year is 8.825.
- Most of the directors are unknown.
- Most of the cast are unknown.
- Most of genre are drama & international movies.
- A huge majority of the releases happened in the last 10-15 years.
- The no.of releases started growing very fast after 2010.
- After 2018 peak,there is a quick decline in the no.of releases in leading into 2020.
- long trail of circles to the left shows that releases before 2005 are rare and considered outliers.
- The highest releases count are in 2018-2019.
- Adult is biggest rating category,so most titles are mature viewers.
- Teen and adult categories have the longest durations.
- Kids content is much shorter than other group.
- Movie length increased sharply from the 1940's to the mid 1960's.
- Some outliers very short (<50 min) and very long (>200 min) films.
- The correlation between release year and duration is -0.21.
- The value is close to 0,so the relationship is not strong
- No strong linear relationship between release year and duration
- Tv show duration are mostly 1-3 seasons